

7-Day Operating System Plan for CDAC C-CAT

This plan is optimized for fast completion, high-weight topics, MCQ solving, numericals, and CDAC-style preparation.
Target: 5–7 focused hours/day.

Day 1 — Processes + Threads + System Calls

- OS basics
- Kernel vs Shell
- Process vs Program
- PCB
- Process States
- Context Switching
- Threads
- fork()
- exec()
- wait()
- Zombie & Orphan process

Practice: 40 MCQs + 10 process-state questions + 10 thread questions

Day 2 — CPU Scheduling (MOST IMPORTANT)

- FCFS
- SJF
- SRTF
- Priority Scheduling
- Round Robin
- Waiting Time
- Turnaround Time
- Response Time
- Starvation
- Aging

Practice: 20 Gantt chart problems + 50 MCQs

Day 3 — Synchronization + Critical Section

- Critical Section Problem
- Race Condition
- Mutex
- Semaphore
- Peterson Solution
- Producer Consumer
- Reader Writer
- Dining Philosopher

Practice: 40 MCQs + scenario-based questions

Day 4 — Deadlocks

- Deadlock conditions
- Resource Allocation Graph
- Safe State
- Banker's Algorithm
- Prevention
- Avoidance
- Detection
- Recovery

Practice: 30 MCQs + 10 Banker problems

Day 5 — Memory Management

- Paging
- Segmentation
- Virtual Memory
- Demand Paging
- Thrashing
- TLB
- FIFO
- LRU
- Optimal Page Replacement

Practice: 20 page replacement problems + 50 MCQs

Day 6 — File System + Disk Scheduling + Linux Basics

- File allocation methods
- Directory structures
- Inodes
- FAT
- FCFS
- SSTF
- SCAN
- C-SCAN
- LOOK
- chmod
- pipes
- redirection

Practice: 40 MCQs + 10 disk scheduling numericals

Day 7 — Full Revision + Mock Test

- Revise formulas
- Deadlocks
- Semaphores
- Page replacement
- Process states
- Scheduling

Practice: 1 Full Mock + 100 mixed MCQs

Most Important Topics for CDAC

Priority	Topic
1	CPU Scheduling
2	Memory Management
3	Deadlocks
4	Synchronization
5	Processes & Threads
6	Disk Scheduling
7	File System
8	Linux Basics

Recommended Resources

- Gate Smashers YouTube Channel
- Jenny's Lectures CS/IT
- GeeksforGeeks OS MCQs
- IndiaBIX OS MCQs
- GATE OS PYQs + CDAC Previous Questions